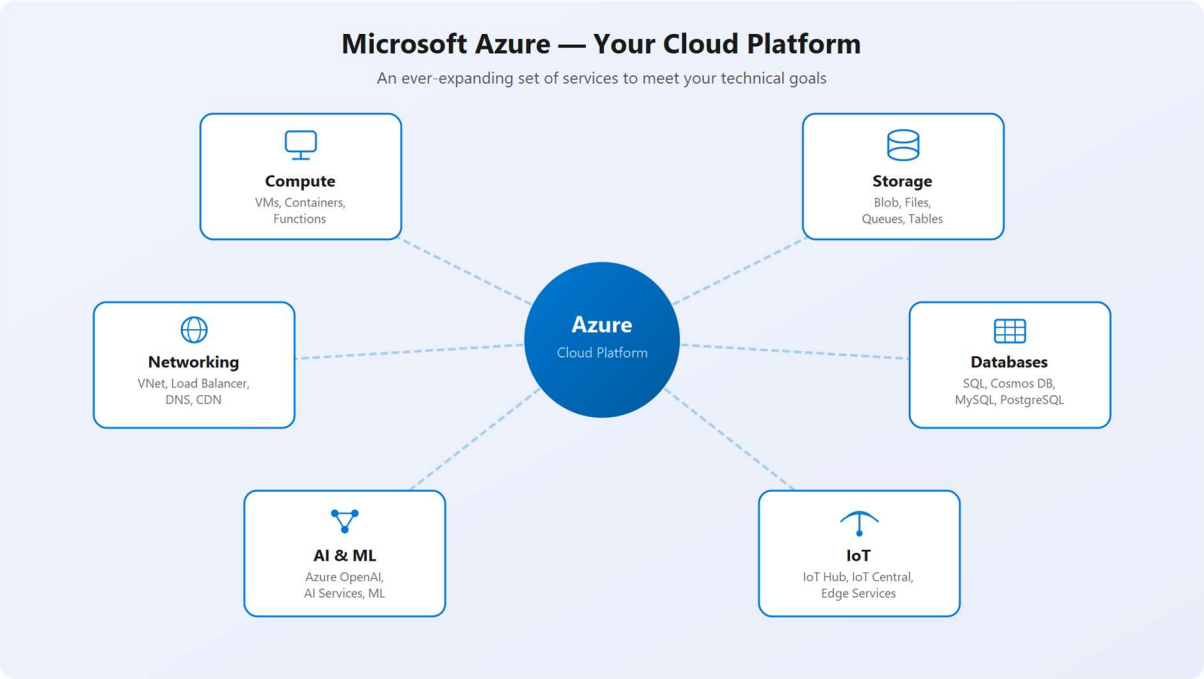
Record of Training

Microsoft Learn: Introduction to Microsoft Azure Fundamentals



Learning Summary

I learned about general cloud concepts and services. Gained knowledge in the different models available for cloud services how each one offers benefits depending on your needs and situation. I also learnt about how cloud services are usually priced and how a consumption-based model benefits the consumer. I also learnt about the shared responsibility model between different platforms such as Infrastructure as a service (**IaaS**), Platform as a service (**PaaS**) and Software as a service (**SaaS)**.

Knowledge Gained

A general understanding of cloud-based systems, what they can offer and how different models can be implemented depending on the consumer needs. I learned the nature of the different

responsibility models and the different cloud models such as **Private, Public and Hybrid.**

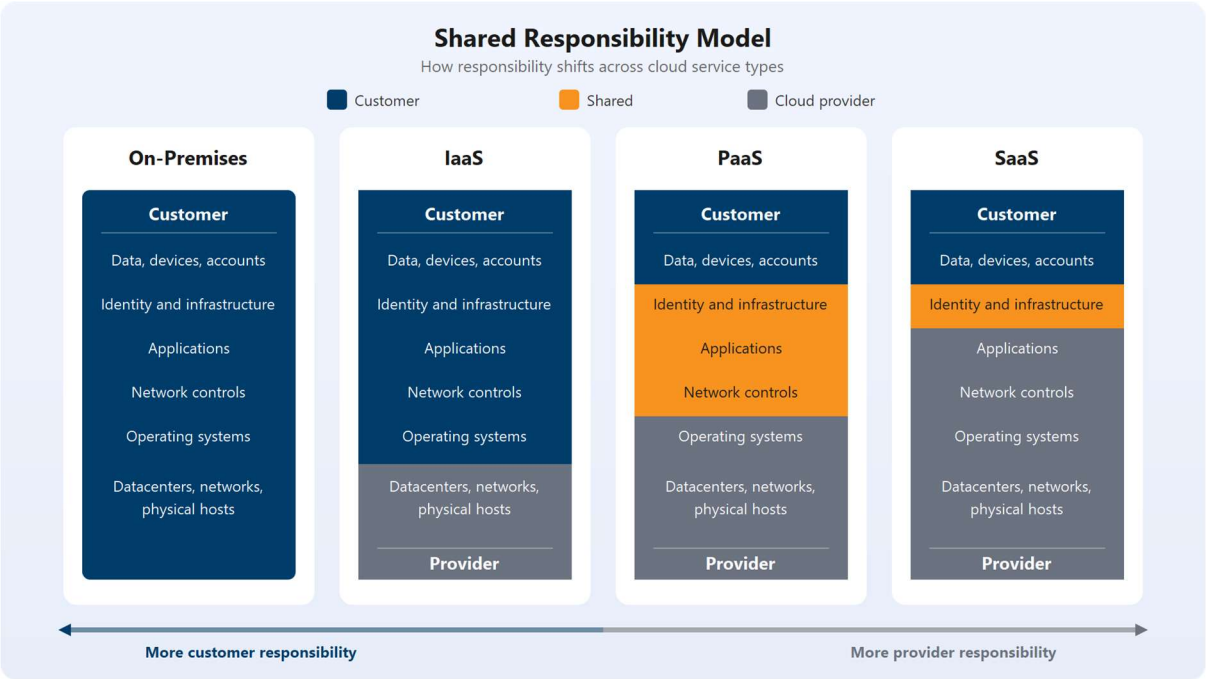
Learning Material

Cloud computing is the delivery of computing services over the internet. They offer services and access to, virtual machines, storage, databases and networking in a non-physical infrastructure. In a specific user case, a team that is suspecting high traffic during launch may offload additional capacity to the cloud instead of buying and securing extra physical servers. They can then scale down accordingly after the launch.

Makes cloud services a more approachable direction for agility and on-demand provisioning.

- Describe the shared responsibility model

There are different responsibility models, **IaaS:** Infrastructure as a service. **PaaS:** Platform as a service. **SaaS:** Software as a service. **IaaS** places most of the responsibility on the consumer with the cloud provider being responsible for the basics of physical security, power and connectivity. On the other end of the spectrum **SaaS** places most of the responsibility on the cloud provider. **PaaS** acts as middle ground between the two, distributing responsibility evenly between cloud provider and consumer.



What always stays with you

- The information and data stored in the cloud
- Devices that are allowed to connect to your cloud (cell phones, computers...)
- The accounts and identities of the people, services and devices within you environment.

What the provider always owns

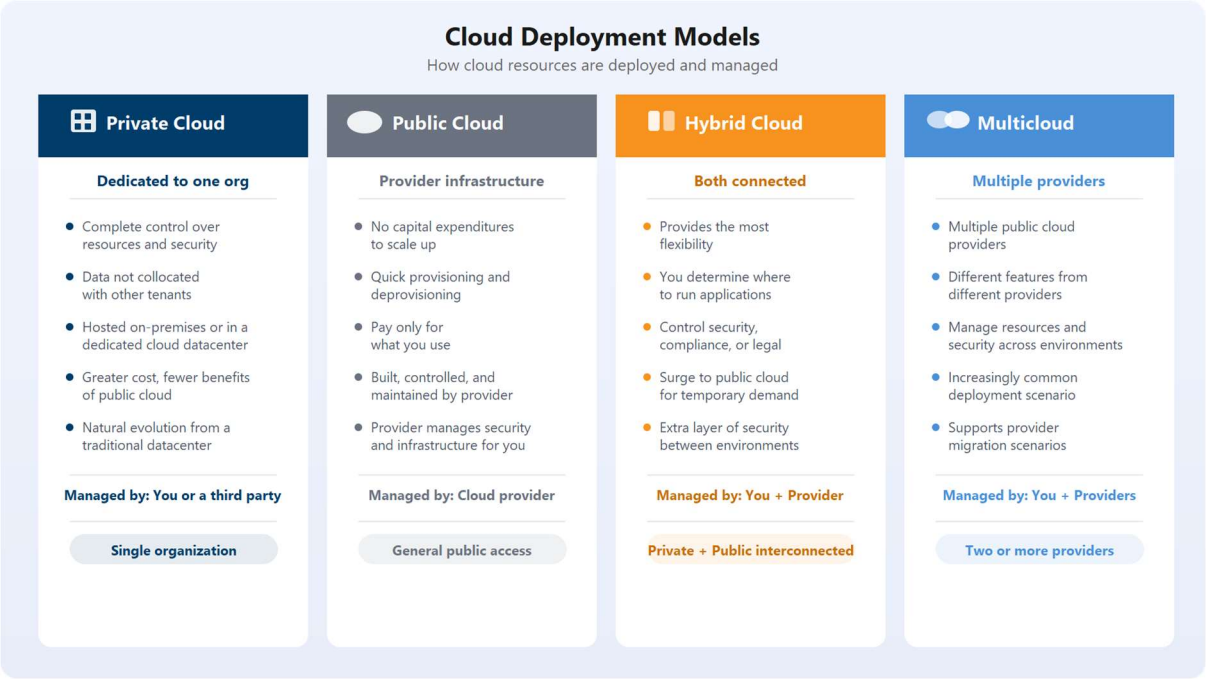
- The physical datacentre
- The physical network
- The physical hosts

What depends on the service type

- Operating systems
- Network controls
- Applications
- Identity and infrastructure

Define Cloud Models

Cloud models define the deployment type of cloud resources. The three main cloud models are: **private, public and hybrid**.



Private Cloud: a cloud system used by a single entity. It offers greater control for your IT team but comes at a greater cost and fewer benefits of the public cloud deployments. Most likely would be hosted from an on-site datacentre or in a dedicated off-site datacentre.

Public Cloud: Built, controlled and maintained by a third-party provider. Anyone can purchase the cloud services, access and use resources.

Hybrid Cloud: Uses both public and private clouds in an inter-connected environment. Can be used to add an extra layer of security, by allowing a private cloud to surge for increased, temporary demand by deploying public cloud resources. Users can decide which services to keep in the public cloud and which to keep to their private cloud.

Multicloud is also an option whether it be temporary while migrating over to another service. Or just that you use different features from different providers for cost or efficiency.

Azure Arc is a set of technologies that helps manage your cloud environment.

Azure VMware Solution lets you run your VMware workloads in Azure with seamless integration and scalability.

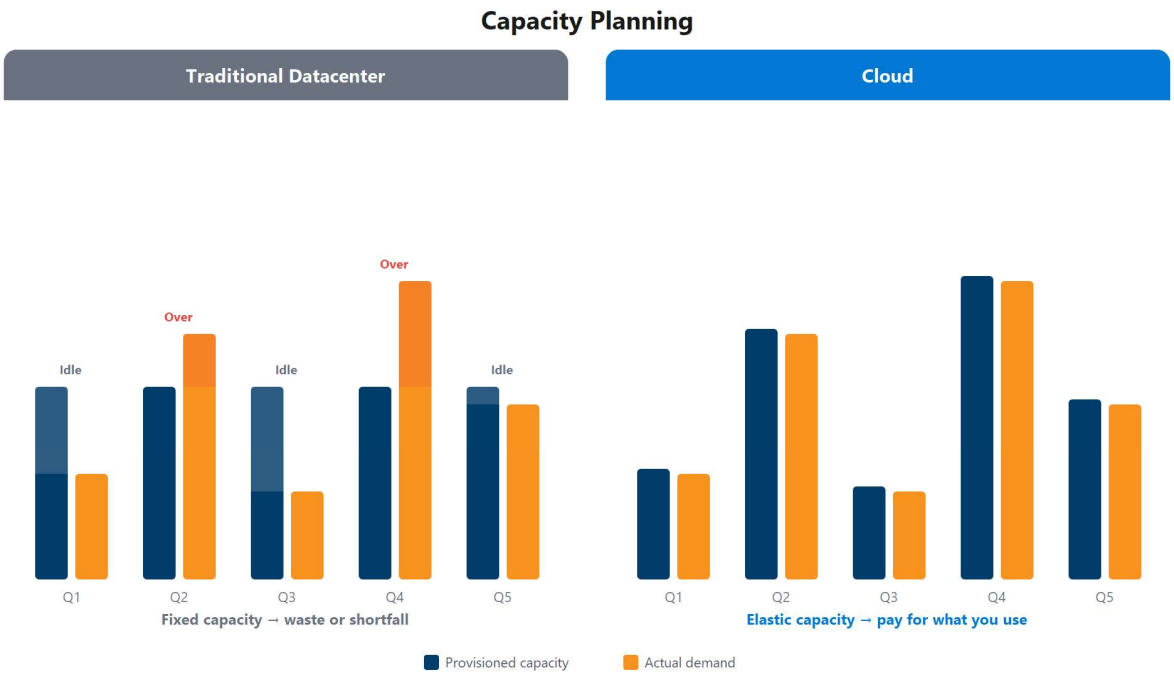
Public cloud	Private cloud	Hybrid cloud
No capital expenditures to scale up	You have complete control over resources and security	Provides the most flexibility
Applications can be quickly provisioned and deprovisioned	Data isn't collocated with other tenants' data	You determine where to run your applications
You pay only for what you use	Hardware must be purchased for startup and maintenance	You control security, compliance, or legal requirements
You don't have complete control over resources and security	You are responsible for hardware maintenance and updates	

The Consumption-Based Model

Basically, you pay for the services you use. Essentially renting compute power, storage and release of those resources when you are done.

This model offers several benefits

- No upfront costs for hardware or datacentre infrastructure
- No need to purchase and manage capacity that may go underutilized
- The ability to add resources when demand increases
- The ability to release resources when demand decreases



Cloud providers offer a pay-as-you-go pricing model. You typically only pay for the services you consume, which helps,

- Plan and manage operating costs
- Run infrastructure more efficiently
- Scale as workload needs change